

The Transmission of Keynesian Supply Shocks

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*The views expressed in this paper are those of the author(s) and do not necessarily represent the views of the Bank of England or its committees.

Keynesian Supply: Intuition

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- ▶ Sectoral supply shocks can lead to aggregate demand deficiencies → Keynesian Supply

Question & Approach

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- [1] Multi-sector panel VAR
- [2] New-Keynesian multi-sector DSGE model

Empirical strategy: Intuition

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 - [1] **Demand:** output and prices move in the same direction in all sectors
 - [2] **Keynesian supply:** output and prices move in opposite direction in at least a few sectors
- ▶ In practice
 - * Identify a shock that moves aggregate output and prices in the same direction in a VAR
 - * Estimate the response of sectoral output and prices to such shock to disentangle [1] or [2]

Literature & Contribution

► Supply shocks and complementarities

- * Guerrieri, Lorenzoni, Straub, Werning (2020)
- * Corsetti, Dedola, Leduc (2008)

► Granular fluctuations & Production networks

- * Gabaix (2011), Gabaix and Koijen (2020)
- * Foerster, Sarte, Watson (2011)
- * Smets, Tielens, Jan Van Hove (2018)
- * Baqaee and Farhi (2020a,2020b)

► Supply vs. demand shocks in the pandemic

- * Brinca, Duarte, Faria-e-Castro (2020)
- * del Rio-Chanona, Mealy, Pichler, Lafond, Farmer (2020)

Data

- ▶ **Data** Sectoral real gross output and gross output deflator
 - * 64 sectors (NAICS 3-digits, ex Oil and Petroleum)
 - * Sample: 2005Q1 to 2019Q1
 - * Source: BEA (<https://apps.bea.gov/iTable/iTable.cfm?ReqID=51&step=1>)
- ▶ **Notation** Sectors are denoted by $i = 1, 2, \dots, N$
 - * Gross real output growth in sector i is y_{it}
 - * Gross output deflator inflation rate in sector i is π_{it}

A Multi-sector factor-augmented panel VAR

- Model sectoral output growth and inflation $x_{it} = (y_{it}, \pi_{it})$ through a factor augmented VAR

$$x_{it} = \Phi_{i0} + \Phi_{i1}x_{it-1} + \Gamma_{if}f_t + u_{it}$$

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- Identification of f_t by aggregation as in Cesa-Bianchi, Pesaran, Rebucci (2020, RFS)
 - * Define $\bar{x}_t$ as the cross-sectional average of x_{it} across sectors
 - * Extract orthogonal innovations e_t from a VAR in $\bar{x}_t$

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- Unobserved factor model can be approximated by:

$$x_{it} = \Psi_{i0} + \Phi_{i1}x_{i,t-1} + \Lambda_{ij}e_t + \sum_{\ell=1}^k \Psi_{i\ell}\bar{x}_{t-\ell} + u_{it}$$

Identification of the common shocks

- Bayesian VAR(1) for aggregate output growth ($\bar{y}_t$) and inflation ($\bar{\pi}_t$)

$$\bar{x}_t = A_0 + \sum_{\ell=1}^k A_{\ell} \bar{x}_{t-\ell} + B e_t$$

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- Identification of e_t with sign restrictions



	Demand-like shock	Supply Shock
Aggr. Output Growth	+	+
Aggr. Inflation	+	-

Identification of the sectoral loadings

- For each draw and associated sign restriction rotation estimate

$$x_{it} = \psi_{i0} + \phi_{i1}x_{i,t-1} + \lambda_i \hat{e}_t + \psi_{i1}\bar{x}_{t-1} + u_{it} \quad \text{for } i = 1, 2, \dots, N$$

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- How do sectoral output and inflation comove in response to demand-like shocks?

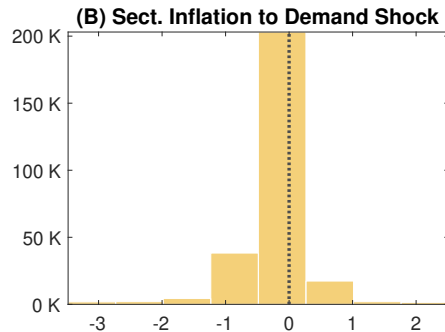
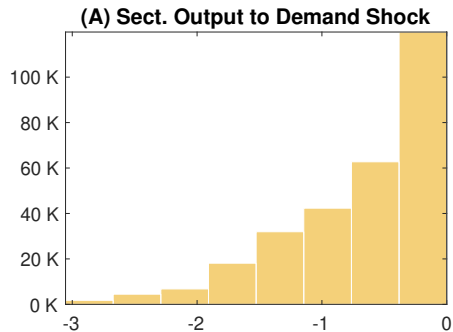
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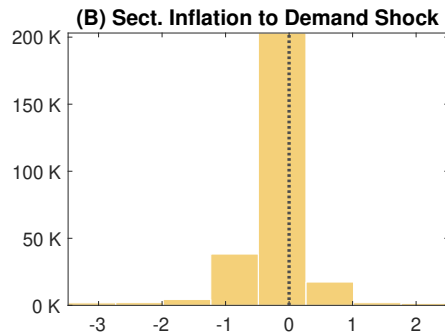
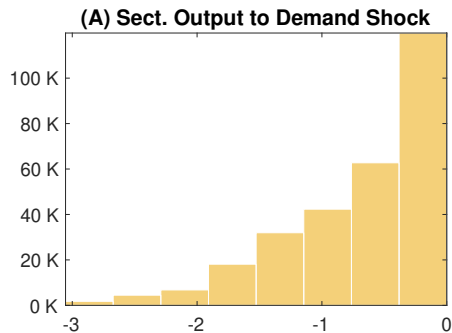
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- How do sectoral output and inflation comove in response to demand-like shocks?
- Plot the distribution of loadings $\lambda_{i,y}$ and $\lambda_{i,\pi}$
 - * Pool loadings across all sectors and draws (64 sectors $\times$ 5,000 draws)
 - * Normalize output loadings $\lambda_{i,y}$ to be negative (account for a few counter-cyclical sectors)
 - * Plot distribution of normalized inflation loadings $\lambda_{i,\pi}$

The distribution of the factor loadings (Λ_i)

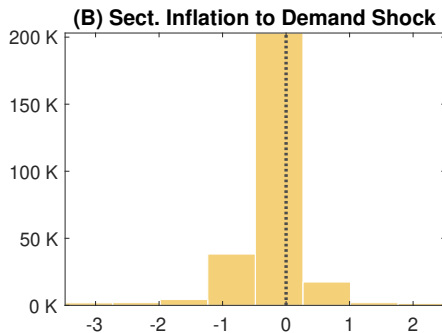
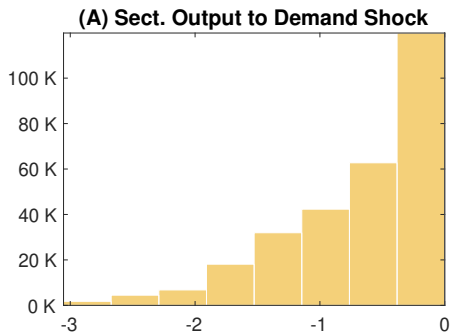


The distribution of the factor loadings (Λ_i)



- ▶ Almost 40% of inflation loadings $\lambda_{i,\pi}$ lie on the 'wrong' side of the distribution
 - * None of the 5,000 draws delivers a demand-like response in all sectors

The distribution of the factor loadings (Λ_i)



- **Interpretation** Evidence is supportive of Keynesian supply transmission mechanism
- * VAR mis-classifies sectoral supply shock with complementarities as aggregate demand shock
 - * Not consistent with a *strict view* of aggregate demand shocks

The distribution of the factor loadings (Λ_i)

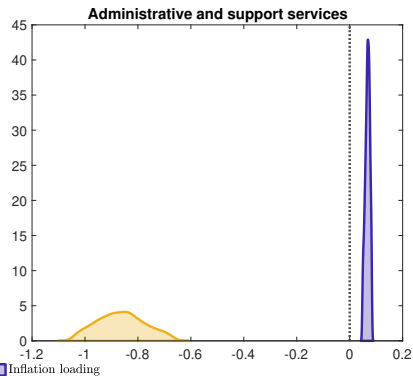
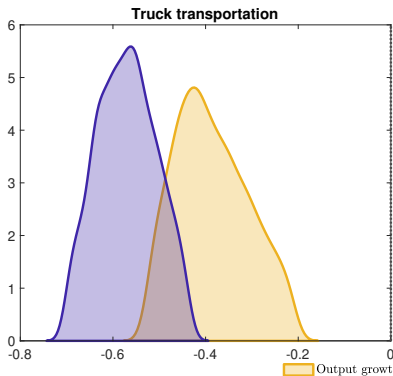
Selected example of loadings to demand shock

- Results are not driven by tail observations in the distribution of sectoral loadings

The distribution of the factor loadings (Λ_i)

Selected example of loadings to demand shock

- ▶ Results are not driven by tail observations in the distribution of sectoral loadings
- ▶ In some sectors, not even one of the 5,000 loadings behaves 'demand-like'



Robustness

- ▶ Adding EBP to aggregate/sectoral VARs [Go](#)
- ▶ Including 2020Q1 [Go](#)
- ▶ Value added instead of gross output [Go](#)
- ▶ Richer dynamics (4 lags) [Go](#)
- ▶ Control for oil shocks [Go](#)
- ▶ Specification in levels [Go](#)

A model-based validation exercise

- ▶ Multi-sector DSGE model with complementarities in consumption and production
[Pasten, Schoenle, Weber (2020); Ghassibe (2020)]
 - * Calibrate the model to the same 64 sectors as in empirical analysis
 - * Obtain simulated sectoral data on gross output and its deflator
 - * Apply our methodology to simulated data

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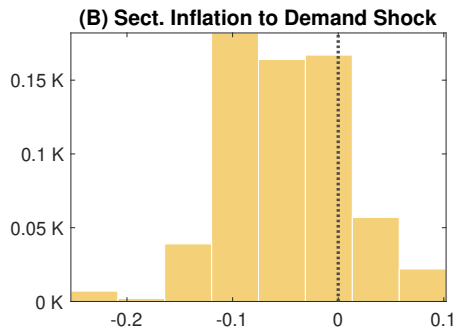
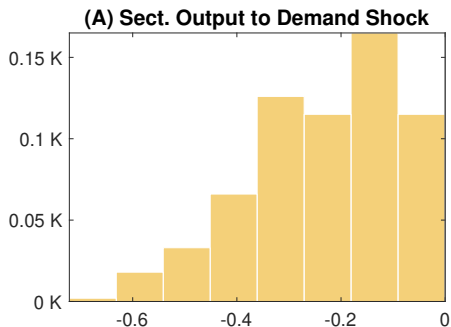
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- ▶ Consider two cases for the exogenous processes driving the model's dynamics
 - [1] Sectoral TFP shocks
 - [2] Aggregate demand shocks

Empirical results based on simulated data

Case [1]: Sectoral TFP shocks

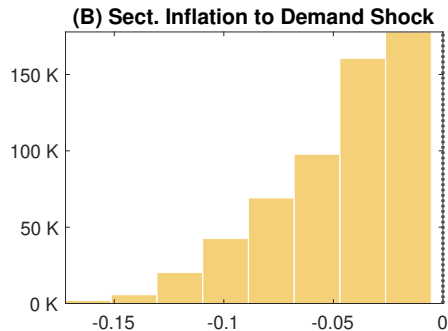
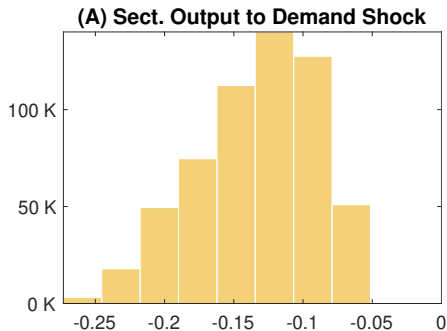
- Distribution of loadings in line with empirical evidence



Empirical results based on simulated data

Case [2]: Demand shocks

- Distribution of loadings inconsistent with empirical evidence



Conclusions

► Main finding

- * Data are supportive of Keynesian supply transmission of sectoral shocks
- * Demand-like shocks derived from aggregate data are contaminated by Keynesian supply shocks

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► Why do we care? Covid-19 → What is the optimal policy response?

- * Tilt the balance of the response in favor of fiscal policy

[Woodford (2020); Guerrieri-Lorenzoni-Straub-Werning (2020)]

Conclusions

► Main finding

- * Data are supportive of Keynesian supply transmission of sectoral shocks
- * Demand-like shocks derived from aggregate data are contaminated by Keynesian supply shocks

► Why do we care? Beyond Covid-19 → What are the sources of business cycle fluctuations?

- * Granular shocks and production networks

[Gabaix (2011); Baqaee-Farhi (2020a,b)]

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► Future research

- * Identify sectoral shocks / DSGE estimation / Cross-country analysis